

no. 30 DATA ETHICS

It is moral principles and guideline that govern how data is collected, processed and analysed and distributed among society and organisations.

1) Cambridge Analytics + Facebook Data 2018

- an app called "this is your digital life" collected quiz data from 270k users. But it also scraped 87m facebook profiles.

2) Geerview AI facial recognition - 2020

Startup Scraped 30B+ photos from facebook, instagram, linkedin, news sites without permission built facial recognition.

3) Amazon's hiring AI hiring tool 2018

Amazon trained an AI Screen (Vs using 10 years of past hiring data).

ISSUE	what went wrong
Consent	- used people's data without clear permission
Transparency	- People were not aware of what was happening
Harm	- manipulated voters, automated bias
Scale	- AI/Data made the unethical behaviour hit millions instantly.

31) Data Governance

It is the process of managing the availability, usability, integrity and security of data in enterprise.

Key components of a good data governance framework
 "rules + roles + tools" this is for, so data isn't a mess and doesn't a mess gets one fixed under GDPR in EU POP11

1) Ownership & Roles - who's accountable

Role	Job
Data owner	- Senior person per domain: "sales data is mine" decides access, quality rules.
Data Steward	- Day-to-day: fixes issues, documents fields, enforces rules
Data Custodian	- IT/engineering: keeps systems secure, runs backups
Data Protection Officer	- GDPR requirement - Owns compliance + privacy.
Data user	- Everyone else. Trained on what they can/can't do.

2) Policies & Standards - The rule Book (clear does everyone follows):

a) Data Quality rules: "customer email must be valid printed phone must have country code"

b) Policy Privacy - what data is collected, why, retention retention time - GDPR requires this

3) Access policy - who can view what, Interns don't get prod customer DB (data base)

mess

170.33 Data Analyst Vs Data Scientist Vs Data Engineer

1) Data analyst - Storyteller

Goal - Turn data into business decisions

2) Data Scientist - The Predictor

Goal - Predict future + test ideas with stats/ML

3) Data engineer - The Plumber

Goal - build reliable data so others can work

Data analyst VS Data Scientist VS Data engineer

Role	Core Question	main output	Time Spent
Data analyst	what happened?	Dashboards, reports, insights	80% SQL / excel BI Tools
Scientist	what will happen + why?	ML models, predictions experiments	60% ph python/ stats, 40% SQL
Data Engineering	How do we get the data	Pipelines, tables, APIs that work	80% python/ spark / SQL infra

no. 34 B1

Business Intelligence = Turning company's data into decisions. It answers "what happened, and why" using so you can act.

function of BI

- 1) Collect data - from all over: Sales system, website
- 2) Clean + combines it into one place: data warehouse
- 3) Visualize it in dash boards / charts people can understand
- 4) Let you ask question - without needing a data scientist

Section E Reflection

no. 35

-I found Data Privacy, Data Ethics and data Governance very interesting. The incident of our facebook-freezing in 2018, it is only now that I understand what happened

no. 36 Most difficult Concept

- 1) Section A
 - 2) Section B
- } most questions on this section's and trying to tackle them was ~~was~~ challenging

No. 37 How do I see data + AI hit career and daily.

On Career - 3 ~~stg~~

Shift	what it means	Example
every job becomes " + AI "	Banks use AI to flag fraud, approve loans	SARS uses data to pre-fill tax returns. AI negotiates your insurance
Discovery vitality tracks steps, gives points		
Data literacy = data = new excel	"Can you read a dash board"? becomes basic like "can you send email"	HR manager checks turnover dash board weekly to spot teams at risk
New jobs created	Prompt engineer, AI auditor, MLOps, data product manager didn't exist 5 years	Banks now hire "model risk analysts" to check AI for BIAS / GDPR